

Temperature Datalogger Calibration Record

Internal calibration record for the single custom PT100 datalogger unit used in the test laboratory.

Instrument Identification

Instrument: PT100 Temperature Datalogger (custom single unit)

Controller: ESP32-S3

RTD Interface: MAX31865

Probe Model: Omega P-L-1/3-1/8-6-0-T-3

Location: Bemidji Test Laboratory

Atmospheric Pressure Source: KBJI METAR (Bemidji Regional Airport, adjacent to site)

Measurement Configuration Snapshot

Item	Value
RTD Type	PT100
Wiring	4-wire
Reference Resistor	430 ohm
MAX31865 Filter	60 Hz
MAX31865 Conversion Mode	PT100 table
Bias Settle Delay	10 ms
Temperature Resolution	0.03125 C nominal
Calibration Model	2-point linear
Calibration Storage	ESP32 NVS

Calibration Procedure

Calibration is performed using two physical reference points: a stirred ice-water slush bath assigned as 0.000 C and a pressure-corrected saturated steam point derived from local atmospheric pressure.

Ice Point Procedure

Distilled water and crushed ice are prepared as a true slurry/slush, not floating ice in bulk water. The probe is immersed approximately 5 to 6 inches into the slush in a 2 L beaker. The probe is gently moved/stirred in the slurry during stabilization to minimize local pockets and maintain equilibrium at the sensing element. This point is assigned a reference temperature of 0.000 C.

Steam Point Procedure

Distilled water is boiled vigorously in a covered beaker. The probe is placed several inches above the water surface within the steam space under the cover, without immersion in liquid water. The steam reference temperature is calculated in firmware from KBJI METAR pressure data by converting the reported altimeter setting to estimated station pressure and then calculating the saturation temperature of water at that pressure.

Model Application

The raw measured temperatures at the two reference points are used to calculate and store a linear 2-point calibration model in nonvolatile memory within the datalogger.

METAR Pressure Record

Field	Record
METAR Station	KBJI - Bemidji Regional Airport
Calibration Date/Time	
METAR Observation Time	
Altimeter Setting (inHg)	
Estimated Station Pressure	
Calculated Steam Temperature (C)	
Notes	Airport is adjacent to calibration site

Calibration Data and Stored Calibration Points

Point	Raw Avg (C)	Reference (C)	Residual (C)
Ice Point	0.471	0.000	-0.471
Steam Point	98.206	97.967	-0.239

Coefficient Record	Value
Point 1 Raw	0.471 C
Point 1 Actual	0.000 C
Point 2 Raw	98.206 C
Point 2 Actual	97.967 C
Calibration Mode	linear

Assigned Uncertainty

Uncertainty Component	Standard Uncertainty (C)
Probe class / interpolation allowance	0.12
Electronics / conversion / quantization	0.03
Calibration realization (ice + steam)	0.10
General use / placement / coupling allowance	0.08

Combined standard uncertainty: 0.173 C

Assigned expanded uncertainty ($k = 2$): **+/-0.30 C** over approximately 0 to 100 C.

Traceability and Use Statement

This calibration is an internally controlled working calibration based on physical temperature reference points. The atmospheric pressure input for the steam point is derived from the adjacent airport METAR record and converted in firmware to estimated station pressure. This record supports internal verification and audit documentation for the single custom datalogger unit. It is not an accredited laboratory calibration certificate.

Authorization

Technician	
Calibration Date	
Next Calibration Due	1 year from calibration date